How-To Library

Informatica

Ingesting Db2 for z/OS Change Data into Snowflake with Cloud Mass Ingestion Databases

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Abstract

This article shows how to set up a Mass Ingestion Databases job to replicate data from a Db2 for z/OS source to a Snowflake Data Cloud target using an incremental load operation.

Supported Versions

- Informatica Intelligent Cloud Services November 2023

Table of Contents

Overview. . . . 2
Change capture mechanism for Db2 for z/OS sources. . . . 3
Task flow. . . . 4
Initial setup of your environment. . . . 5
Step 1: Verify that your source and target versions are supported. . . . 5
Step 2: Check Mass Ingestion Databases system requirements. . . . 5
Step 3: Set up an organization and organization users. . . . 5
Step 4: Download and install a Secure Agent. . . . 6
Step 5: Configure the Database Ingestion service for your runtime environment. . . . 9
Creating projects and project folders. . . . 11
Db2 for z/OS sources. . . . 11
Installing and configuring the stored procedure for Db2 for z/OS CDC. . . . 12
Db2 for z/OS privileges. . . . 16
Snowflake targets. . . . 18
Configuring connections. . . . 19
Db2 for zOS Database Ingestion connection properties. . . . 19
Snowflake Data Cloud connection properties. . . . 20
Configuring a database ingestion task. . . . 25
Step 1: Defining basic task information. . . . 25
Step 2: Configuring the Db2 for z/OS source. . . . 26
Step 3: Configuring the Snowflake target. . . . 31
Step 4: Configuring runtime options. . . . 34
Deploying the database ingestion task. . . . 36
Running the deployed database ingestion job. . . . 36
Monitoring database ingestion jobs. . . . 36
Other resources. . . . 37

Overview

This article describes how to implement a database ingestion incremental load task that replicates change data from a Db2 for z/OS source to a Snowflake Data Cloud target. It covers initial setup tasks for the Mass Ingestion environment and how to configure, deploy, run, and monitor a database ingestion task.



The use cases for replicating real-time change data from a mission-critical Db2 for z/OS production system to a Snowflake data warehouse include:

- Minimizing intrusive, non-critical work, such as offline reporting or analytical operations on the Db2 for z/OS source database system.
- Reducing costly CPU usage, disk space, and CDC processing time on the Db2 for z/OS source database system.
- Getting the latest database changes to your data warehouse in real time for downstream processing.
- Migrating to a subscription-based Cloud environment.

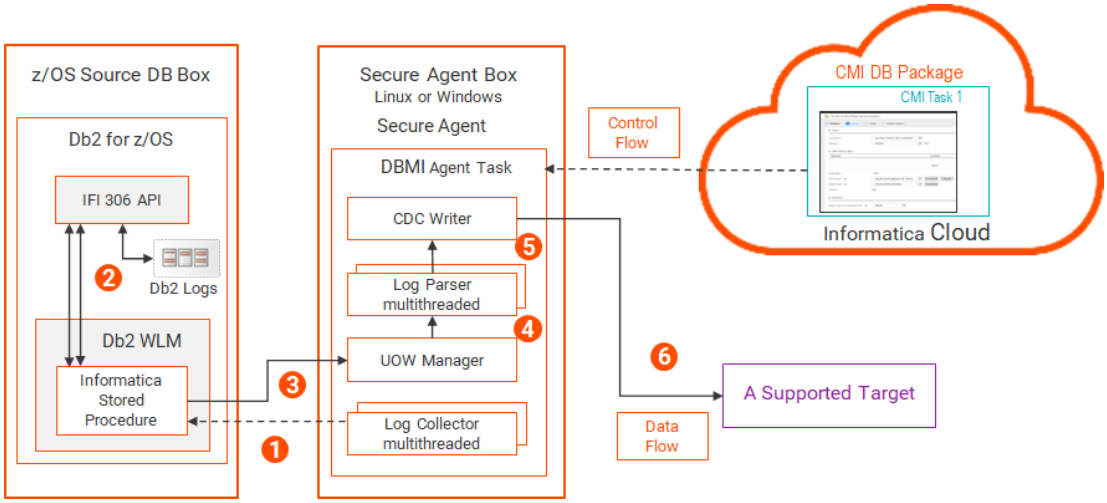
For more information, see the Cloud Mass Ingestion documentation on the Doc Portal at https://docs.informatica.com/integration-cloud/cloud-mass-ingestion/current-version.html.

Change capture mechanism for Db2 for z/OS sources

Mass Ingestion Databases provides a single change capture mechanism and architecture for capturing change data from a Db2 source on z/OS and applying the data to a target. This architecture uses multithreaded processing to optimize the performance of collecting data and of parsing and transforming the data into a format the target accepts.

The Secure Agent must run on a Linux or Windows box, separate from the Db2 z/OS source system. The network bandwidth between the z/OS system and the Secure Agent box must be robust. Informatica recommends network transfer rates that can handle100s of gigabytes or one or more gigabits of log data. If the network transfer rate is not capable of delivering the log data to the Secure Agent at speeds equal to or greater than the rate at which Db2 produces log data of CDC interest, the database ingestion job will not be able to provide the data to the target in a timely manner. If the data throughput does not meet your SLA, consider changing the hardware to increase the ethernet bandwidth between the z/OS system and the Secure Agent box.

The following image shows the Db2 for z/OS change capture components and data flow:



1. The multithreaded log collector, which runs in the DBMI Agent service under the Secure Agent, issues multiple concurrent requests for Db2 log data to the Db2 stored procedure.
   Each request includes a starting RBA, or a starting LRSN if the logs are in a data-sharing environment, along with a list of tables of CDC interest for database ingestion.

   While processing a sequence of log data, the log collector can request the next sequence of log data.
2. The Db2 Instrumentation Facility Interface (IFI) API extracts data for the selected source tables of CDC interest from active and archive logs. The IFI then transfers the data, in raw, native form, to the Informatica Db2 stored procedure in z/OS Workload Manager (WLM).
3. The Db2 stored procedure returns the UOWs with captured data to the UOW Manager.
4. The UOW Manager sends the UOWs in commit order to the log parser.
5. The multithreaded log parser concurrently parses DML changes from the committed UOWs. The result is raw native Db2 log data in the format that the target type expects.

   Note: Because this resource-intensive activity occurs on the Secure Agent box, CPU consumption on the z/OS system is minimized.
6. The CDC writer applies the formatted data to the target.

Task flow

Perform the following tasks, in the order shown, to implement Db2 for z/OS to Snowflake incremental load processing.

1. "Initial setup of your environment" on page 5.
2. "Creating projects and project folders" on page 11 in which to store your database ingestion tasks.
3. Prepare and learn about "Db2 for z/OS sources" on page 11 and "Snowflake targets" on page 18.
4. "Configuring connections" on page 19 if not already available.
5. "Configuring a database ingestion task" on page 25.
6. "Deploying the database ingestion task" on page 36.
7. "Running the deployed database ingestion job" on page 36.
8. "Monitoring database ingestion jobs" on page 36.

Initial setup of your environment

You must complete some initial setup tasks in your Informatica Intelligent Cloud Services environment before you start configuring connections and database ingestion tasks.

1. Verify that your source and target versions are supported.
2. Check that you system meets the minimum system requirements.
3. Create an organization and add users to it.
4. Download and install a Secure Agent.
5. Configure the Database Ingestion agent service for your runtime environment.

Step 1: Verify that your source and target versions are supported

To determine the source and target types and versions that are supported for each ingestion type, check the KB article What are the supported sources and targets for IICS Cloud Mass Ingestion service?.

To determine the web browsers that are compatible with Mass Ingestion and the operating systems that are supported for the Secure Agent, check the Product Availability Matrix (PAM) for Informatica Intelligent Cloud Services (IICS) on the Informatica Network at https://network.informatica.com/community/informatica-network/product-availability-matrices/.

Step 2: Check Mass Ingestion Databases system requirements

The following table lists the Mass Ingestion Databases minimum system requirements for the Secure Agent:

Component	Minimum requirement
Cores per CPU	8 minimum, 16 recommended if you need to process a large number of source tables in an initial load
Memory	32 GB
Disk space	5 GB per job, based on a row size of 2 KB

Step 3: Set up an organization and organization users

An organization is a secure area within the Informatica Intelligent Cloud Services repository that stores your licenses, user accounts, ingestion tasks, and information about jobs and security. Set up an organization in Administrator.

If your organization has the Organization Hierarchy license, you can also create one or more sub-organizations within your organization. You can create sub-organizations to represent different business environments in your company. For example, you might create separate sub-organizations to represent your development, testing, and production environments.

Perform the following tasks:

- On the **Organization** page, configure organization properties such as the organization name and address, authentication information, and notification email addresses. Click the Sub-organizations tab to add a sub-organization to the organization.
- On the **Users** and **User Groups** pages, add and configure users and user groups for the organization. On the User Roles page, you can configure user role permissions, which can be assigned to users and user groups.

Administrator provides context-sensitive help for all of these pages. To display the help for the page in which you're working, click the help (?) icon and select **Online Help**.

Step 4: Download and install a Secure Agent

A Secure Agent is a lightweight program that runs tasks and enables secure communication across the firewall between your organization and Informatica Intelligent Cloud Services.

Install the Secure Agent on all the systems from which you want to use Mass Ingestion Databases. You can install a Secure Agent on Windows or Linux.

When you download and install a Secure Agent, a *Secure Agent group*, also called a runtime environment, is created. A Secure Agent group can contain one Secure Agent or multiple agents.

Note: Mass Ingestion Databases does not support the Hosted Agent or serverless runtime environment.

Downloading and installing the Secure Agent on Linux

To install the Secure Agent on a Linux machine, download and run the Secure Agent installer and then register the agent.

Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine using the same Linux user account. If any other Secure Agent exists, you must uninstall it first.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Runtime Installer**.
3. Select **Secure Agent** as the Environment Type.
4. Select **Linux 64** as the Platform.
5. Click **Copy** to copy the install token.

   You will need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 5 to obtain a new token. You don't need to download the installer again.
6. Click **Download**.

   The installer is downloaded to your machine. The name of the installer is `agent64_install_ng_ext.<agent core version>.bin`.
7. Save the installer to a directory on the machine where you want to run the Secure Agent.

   Note: Ensure that the file path doesn't contain spaces or multibyte characters. If the file path contains spaces, the installation might fail. If the path contains multibyte characters, the Secure Agent might not start.
8. From a shell command line, navigate to the directory where you downloaded the installation program and enter the following command:

   ```
   ./agent64_install_ng_ext.bin -i console
   ```
9. When the installer completes, navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore
   ```
10. To start the Secure Agent, enter the following command:

    ```
    ./infaagent.sh startup
    ```

    The Secure Agent Manager starts. Register the agent using the user name that you use to access Informatica Intelligent Cloud Services and also supply the install token.

11. To register the agent, in the `<Secure Agent installation directory>/apps/agentcore` directory, enter one of the following commands using your Informatica Intelligent Cloud Services user name and the token that you copied:

    - To add the agent to its own Secure Agent group, use the following command:

      ```
      ./consoleAgentManager.sh configureToken <user name> <install token>
      ```

    - To add the agent to an existing Secure Agent group, use the following command:

      ```
      ./consoleAgentManager.sh configureTokenWithRuntime <user name> <install token>
      <Secure Agent group name>
      ```

      Note: If the command includes a Secure Agent group name that doesn't exist, the Secure Agent is not assigned to a group. Be sure to use a valid Secure Agent group name.

    The following table lists the command options:

Option	Description
User Name	Required. Informatica Intelligent Cloud Services user name of the user installing the Secure Agent.
Install Token	Required. The install token that you copied when you downloaded the runtime installer.
Secure Agent group name	Optional. Include a group name when you want to add the agent to an existing Secure Agent group instead. If this option isn't included in the command, the agent will be in its own Secure Agent group.

    You can check the registration status of a Secure Agent using the following command:

    ```
    ./consoleAgentManager.sh isConfigured
    ```

Downloading and installing the Secure Agent on Windows

To install the Secure Agent on a Windows machine, you must download and run the Secure Agent installation program and then register the agent.

Secure Agent registration requires an install token. To get the install token, copy the token when you download the agent or use the **Generate Install Token** option in Administrator. The token expires after 24 hours.

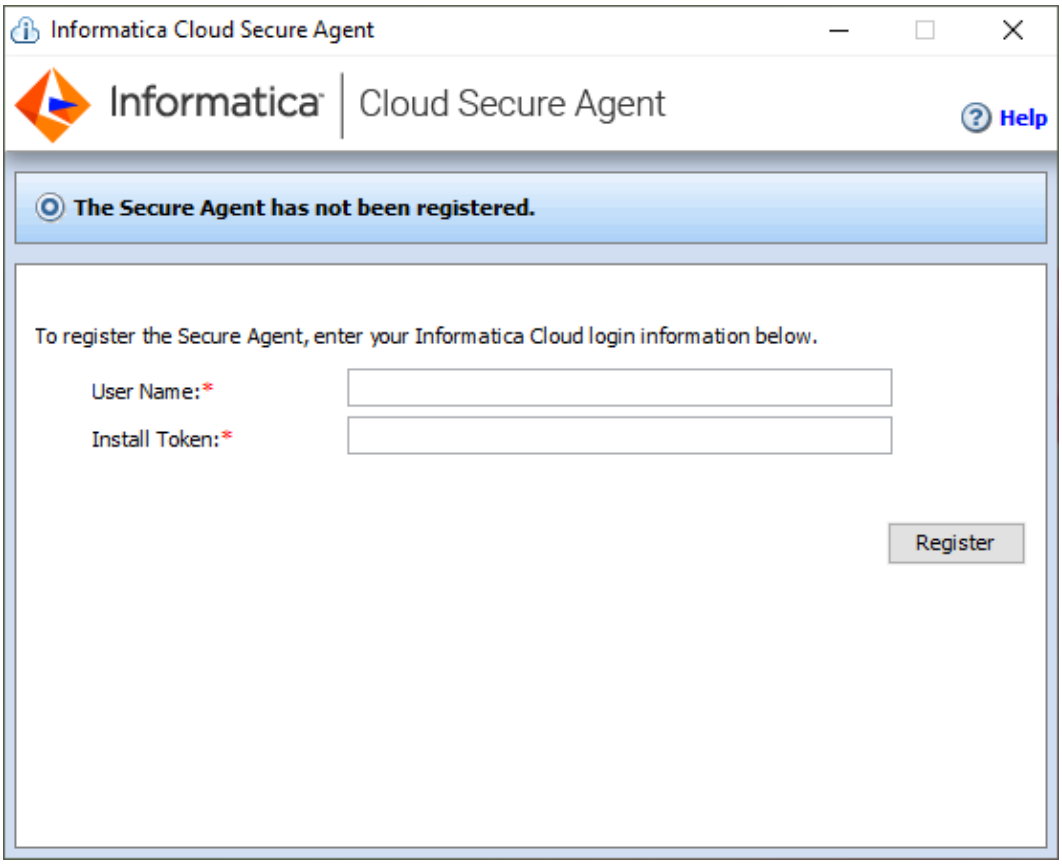
Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine. If there is, you must uninstall it.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Secure Agent**.
3. Select the Windows 64-bit operating system platform, copy the install token, and then click **Download**.

   The installation program is downloaded to your machine. The name of the installation program is `agent64_install_ng_ext.<agent core version>.exe`.
4. Run the installation program as an Administrator:
   a. Specify the Secure Agent installation directory, and click **Next**.
   b. Click **Install** to install the agent.

The Secure Agent Manager opens and prompts you to register the agent as shown in the following image:



5. If you did not copy the install token when you downloaded the agent, click **Generate Install Token** on the **Runtime Environments** page in Administrator, and copy the token.
6. In the Secure Agent Manager, enter the following information, and then click **Register**:

Option	Description
User Name	User name that you use to access Informatica Intelligent Cloud Services.
Install Token	Token that you copied.

The Secure Agent Manager displays the status of the Secure Agent. It takes a minute for all of the services to start.

7. If your organization uses an outgoing proxy server to connect to the internet, enter the proxy server information.
8. Close the Secure Agent Manager.

The Secure Agent Manager minimizes to the taskbar and continues to run as a service until stopped.

Step 5: Configure the Database Ingestion service for your runtime environment

Mass Ingestion Databases uses the Database Ingestion agent service to run database ingestion jobs. The Database Ingestion service is automatically downloaded to the system when you download the Secure Agent.

After you download the Secure Agent to your runtime environment and enable the Database Ingestion service, the Database Ingestion packages are pushed to the on-premises system where the Secure Agent runs. You can then optionally configure properties for the Database Ingestion service that runs on the Secure Agent.

Database Ingestion service properties

To change or optimize the behavior of the Database Ingestion service that your Secure Agent group uses, configure Database Ingestion properties for your runtime environment.

To configure the properties, open your runtime environment and click **Edit**. Under **System Configuration Details**, select the **Database Ingestion** service and the **DBMI_AGENT_CONFIG** type.

The following table describes the Database Ingestion agent service properties:

Property	Description
maxTaskUnits	The maximum number of database ingestion task units that can run concurrently on an on-premises machine where the Secure Agent is running. Task units are not related to the capacity and availability of your hardware or software. You can configure maxTaskUnits to precisely control CPU usage. Valid values are 1 to 2000000000 (2 billion). To calculate a reasonable number of task units for your Secure Agent machine, Informatica recommends that you divide the number of cores by 3 or 4. For example, if you have an 8-core machine, you could set this property to 2. Then monitor CPU usage and adjust the property value as needed to tune performance. During initial load processing, this property determines the number of tables that can be unloaded simultaneously. Remaining tables are queued and start unload processing when resources become available. **Note:** A single job can process many tables. The total number of tables that can be processed is limited only by available memory. On the average, 25 MB of RAM is required per table for an initial load task based on a 1 KB row size. During incremental load processing, this property determines the number of database ingestion jobs that can run simultaneously. Setting this property to a value greater than the number of cores on the Secure Agent machine can increase parallelism for task execution but also cause performance bottlenecks at task execution time.
serviceLogRetentionPeriod	The number of days to retain each internal Database Ingestion service log file after the last update is written to the file. When this retention period elapses, the log file is deleted. The default value is 7 days. Service logs are retained on the Secure Agent host where they are created: `<infaagent>/apps/Database_Ingestion/logs`.

Property	Description
taskLogRetentionPeriod	The number of days to retain each job log file after the last update is written to the file. When this retention period elapses, the log file is deleted. The default value is 7 days.
serviceUrl	The URL that the Database Ingestion service uses to connect to the Informatica Intelligent Cloud Services cloud.
logLevel	The level of detail to include in the logs that the Database Ingestion service produces. Options are: - TRACE - DEBUG - INFO - WARN - ERROR The default value is TRACE.
taskExecutionHeapSize	The maximum heap size, in gigabytes, for the Task Execution service. This value, in conjunction with maxTaskUnits property, affects the number of concurrent database ingestion tasks that can run on a Secure Agent. Try increasing the heap size to run more tasks concurrently. Enter this value followed by "g" for gigabytes, for example, '9g'. The default value is '8g'.
useProxy	Set this property to true to enable the DBMI Agent to go through a proxy when connecting to or writing data to targets. The DBMI Agent then uses the proxy settings from the Secure Agent proxy configuration. By default, proxy settings are not used.
intermediateStorageDirectory	For incremental load and combined initial and incremental load jobs, the local root directory under which intermediate files that contain data are stored when the **Enable Persistent Storage** option is selected in the associated task definitions.
storageBackupDirectory	For incremental load and combined initial and incremental load jobs, the path to the directory that stores backup files when the **Enable Persistent Storage** option is selected in the associated task definitions.
storageProperties	For incremental load and combined initial and incremental load jobs, a comma-separated list of key=value pairs that is used when the **Enable Persistent Storage** option is selected in the associated task definitions. Specify this property only at the direction of Informatica Global Customer Support.
task_container.jvm.allowExceptionForInvalidEncodedData	If you receive transliteration errors that report invalid encoding to UTF-8, and you do not want to repair or correct the source data, set this property to false so that database ingestion jobs do not fail when trying to unload the data from the source. With this setting, the Database Ingestion service passes an equivalent Java property to the DataDirect JDBC driver to prevent the exception from occurring. After you set this property, you must restart the Database Ingestion service.

Database Ingestion service environment variable

To change or optimize the behavior of the Database Ingestion Agent, define the following environment variable:

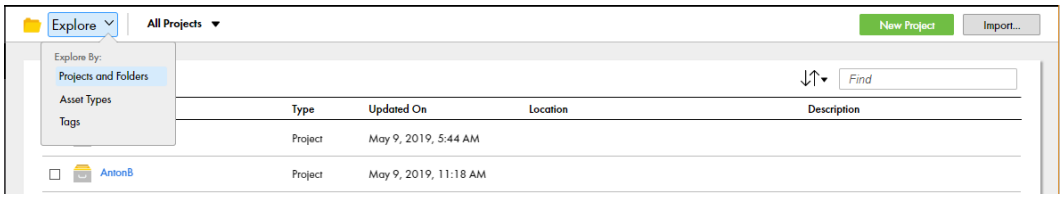
Environment Variable	Description
DBMI_WRITER_CONN_POOL_SIZE	Indicates the number of connections that a database ingestion job uses to propagate the change data to the target. The default value is 8. Valid values are 4 through 8.

Note: After you define or change an environment variable, restart the Database Ingestion Agent for the changes to take effect.

Creating projects and project folders

Create a project to contain your ingestion task assets. You can create one or more folders under the project to logically organize your assets. However, you cannot create subfolders under a folder.

1. In the Mass Ingestion service, open the **Explore** page.
2. If the **Explore** page shows objects other than projects, select **Projects and Folders** in the **Explore** menu.



3. To create a project, click **New Project**.
4. In the **New Project Properties** dialog box, enter a project name up to 255 characters in length. You can also enter an optional description of the project.

   A project name cannot contain the following characters: `#?`|{}"^&[]/\.`
5. If you want to add a folder under the project, select the project and click **New Folder**.
6. In the **New Folder Properties** dialog box, enter a folder name up to 255 characters in length. You can also enter an optional description of the project.

   A folder name cannot contain the following characters: `#?`|{}"^&[]/\.`

When you define an ingestion task, you must specify the project or project folder location to contain the task definition.

Db2 for z/OS sources

To use Db2 for z/OS sources in database ingestion tasks, first prepare the source database and review the usage considerations.

Source preparation

- Verify that you use a Db2 for z/OS version that Mass Ingestion Databases supports. See the KB article FAQ: What are the supported sources and targets for IICS Cloud Mass Ingestion service?.
- Database ingestion incremental load jobs with a Db2 for z/OS source use a stored procedure to call the Db2 Instrumentation Facility Interface (IFI) to read change data from Db2 logs on the z/OS source system. Mass Ingestion Databases delivers the stored procedure libraries and JCL in the Db2 for zOS Database Ingestion connector package ZIP file. You must receive the stored procedure libraries into an APF-authorized library and

then customize the JCL for your environment. The stored procedure runs in a Workload Manager (WLM) address space on the Db2 source system. For more information about z/OS system requirements, stored procedure setup, and required permissions, see "Installing and configuring the stored procedure for Db2 for z/OS CDC" on page 12.

- When you define a Db2 source in a database ingestion incremental load task, you must select the option **Enable CDC for all columns** in the **CDC Script** field. Mass Ingestion Databases generates a script for enabling Db2 DATA CAPTURE CHANGES on the source tables and on certain Db2 catalog tables that are used for CDC. After this attribute is set for one job, all other jobs recognize that the attribute is enabled in Db2 for the catalog tables needed. You can either execute the CDC script from the user interface if you have sufficient privileges or ask a Db2 DBA who has SYSDBA authority to run the script.
- If you plan to perform incremental load processing, grant all required user privileges. For more information, see "Db2 for z/OS privileges" on page 16.
- Mass Ingestion Databases uses the Progress DataDirect JDBC IBM DB2 driver to connect to a Db2 for z/OS source. For a new implementation of Mass Ingestion Databases with Db2 for z/OS sources, the user who first runs an incremental job must have SYSADM or SYSDBA authority to establish JDBC connectivity.

Usage considerations

- Because Mass Ingestion Databases expects each source table row to be unique, Informatica recommends that each source table have a primary key. Mass Ingestion Databases does not honor unique indexes in place of a primary key. If no primary key is specified, Mass Ingestion Databases treats all columns as if they are part of the primary key.
- The default setting for z/OS is that schema drift is enabled when you run the ALTER statements for the catalog tables. Once schema drift is enabled for one job, it is enabled for all jobs. If jobs are running when the ALTER statements are running, the running jobs do not have schema drift enabled until they are subsequently stopped and resumed.
- Mass Ingestion Databases does not support schema drift for Db2 11 for z/OS sources.
- Mass Ingestion Databases does not support the following Db2 for z/OS data types:
  - BLOB
  - CLOB
  - DBCLOB
  - XML

  Database ingestion jobs propagate nulls for columns that have these data types.

Installing and configuring the stored procedure for Db2 for z/OS CDC

To perform Db2 for z/OS CDC processing for incremental load jobs, Mass Ingestion Databases provides a stored procedure that runs on the z/OS source system. The stored procedure calls the Db2 Instrumentation Facility Interface (IFI) to collect change data from the Db2 logs.

z/OS system requirements

Before you begin, verify that the Db2 for z/OS source system meets the following requirements:

- The z/OS system has the recommended operating system version of 2.3 or later.
- The Db2 for z/OS source uses Db2 version 11 or 12.
- Ensure that a Workload Manager (WLM) address space exists for executing the Mass Ingestion Databases stored procedure.

If you have not set up a Db2 WLM address space, see the following IBM documentation for more information:

- For Db2 11:
  https://www.ibm.com/support/knowledgecenter/en/SSEPEK_11.0.0/inst/src/tpc/db2z_setupwlmenvironment.html
- For Db2 12:
  https://www.ibm.com/support/knowledgecenter/en/SSEPEK_12.0.0/inst/src/tpc/db2z_setupwlmenvironment.html

- Ensure that the library where the stored procedure for Db2 for z/OS CDC will be received exists and is APF-authorized on the z/OS system.

Install the Stored Procedure Libraries and Customize the JCL

On your client machine, perform the following steps:

1. Verify that the Db2 for zOS Database Ingestion connector is available.

   The Db2 for zOS Database Ingestion connector package contains the Db2 for z/OS stored procedure libraries. When the connector is enabled for the Secure Agent Group (runtime environment), the connector package .zip file is downloaded to the downloads folder in the installation location. The package name has the format package-DB2ZMI.*nnnnn*, where *nnnnn* is an incremented package version number. If multiple package versions exist, use the latest one.
2. Unzip the package-DB2ZMI.*nnnnn*.zip file. The stored procedure files are added to the Db2WLMStoredProcedure folder under the package name.
3. Use FTP to transfer the #STPINST file in the Db2WLMStoredProcedure folder to a sequential file, PDS, or PDSE on the z/OS system.

   Notes:

   - Transfer the file without binary mode set.
   - Add a High Level Qualifier (HLQ) if needed to meet system requirements.
4. Use FTP or another file transfer method to transfer each of the following files to a separate data set on the z/OS system:

   - DBMI.ZOS.DBRMLIB.XMI
   - DBMI.ZOS.LOADLIB.XMI
   - DBMI.ZOS.USERJCL.XMI

   Notes:

   - Add or edit the HLQ if needed to meet system requirements.
   - Transfer the files in binary mode.
   - Ensure that each data set has the DCB attributes of LRECL=80, BLKSIZE=3120, and RECFM=FB.

     You might need to pre-allocate a data set to specify the required attribute values.

On the z/OS system, use TSO to receive the transmit (XMI) data sets into an APF-authorized library and then edit the stored procedure JCL member. Also, set Db2 user privileges and add a row to the resource limit table if used.

1. Receive the DBRMLIB transmit data set:

   ```
   RECEIVE INDATASET(DBMI.ZOS.DBRMLIB.XMI)
   ```

   Notes:

   - If you specified a HLQ when transferring the data set to z/OS, include the HLQ.

- When you see the message `INMR906A Enter restore parameters or 'DELETE' or 'END' +`, enter your APF-authorized library:

  ```
  DA(your.library_name) UNIT(unit) VOLUME(volume)
  ```

  The UNIT() and VOLUME() operands are optional. Include them if your installation does not put RECEIVE files on a work unit or volume by default.

2. Receive the LOADLIB transmit data set:

   ```
   RECEIVE INDATASET(DBMI.ZOS.LOADLIB.XMI)
   ```

   See the Notes in Step 1.

3. Receive the USERJCL transmit data set:

   ```
   RECEIVE INDATASET(DBMI.ZOS.USERJCL.XMI)
   ```

   See the Notes in Step 1.

4. Customize the #STPINST file that contains the stored procedure JCL for your environment.

   The JCL creates the stored procedure and a global temporary table that holds the results of the requests to the Db2 IFI for data. It also binds the stored procedure package.

   Notes:

   - Based on the comments at the top of the JCL file, replace variables in the JCL with the values appropriate for your z/OS environment, including the Db2 subsystem name (!DSN!), stored procedure schema name (!SCHEMA!), stored procedure name (!STRPRC!), WLM environment name (!WLMENV!), and the name of the DBRMLIB transmit data set (!DBRMLIB! received in Step 1.
   - If you used a HLQ for the received data sets, include the HLQ in the JCL.
   - The WLM environment name is specified in the procedure APPLENV parameter or in the EXEC PARM of the WLM address space.
     In the procedure parameter:

     ```
     //STARTING EXEC DSNBWLMG,DB2SSN=DSNB,APPLENV='DSNBWLM_GENERAL'
     ```

     In the EXEC PARM:

     ```
     PARM='DSNB,40,DSNBWLM_GENERAL'
     ```

   - You can use the received LOADLIB library after it is APF-authorized, or copy the contents of the library to your own APF-authorized library.
   - The STEPLIB concatenation in the WLM address space must contain only APF-authorized libraries for the Db2 IFI to run.

5. Customize the JCL in members of the received USERJCL data set. For more information, see " Db2 for z/OS USERJCL Data Set " on page 14.

6. Ensure that the required Db2 privileges have been granted before you run the stored procedure JCL job. For more information, see "Db2 for z/OS privileges" on page 16.

7. Add a row in the Db2 DSNRLST*xx* resource limit table for database ingestion packages to ensure processor resources are adequate for stored procedure processing. Otherwise, incremental load jobs might abend. Add the row with the following columns:

   - An AUTHID column with the authentication ID that the database ingestion task uses OR an RLFPDG column with the same package name as the database ingestion task OR both of these columns.
   - An ASUTIME column defined with NULL or with a resource limit that is greater than the default limit.

   Then issue the Db2 -START RLIMIT command for For the changes to the resource limit table to take effect.

Db2 for z/OS USERJCL Data Set

The downloaded USERJCL data set is a partitioned data set (PDS) or extended partitioned data set (PDSE) that contains a JCL member for running a job that collects information that is not readily available otherwise when using Informatica-supplied WLM Db2 stored procedure.

You can fully customize member JCL in the USERJCL PDS or PDSE. Informatica recommends that you create a copy of a delivered JCL member under another name and then customize the copy so that original member is retained intact for reference.

If you install multiple stored procedures on multiple Db2 subsystems, you can install only one USERJCL library and create members in it that are tailored for each Db2 subsystem. Alternatively, you can create a separate library for a specific Db2 subsystem. Make sure that the database ingestion job contains the correct library and member information for the Db2 subsystem.

The USERJCL PDS or PDSE contains the following member:

LOGINV member

LOGINV contains JCL for a job that obtains a Db2 log inventory list. If you set the **Initial Start Point for Incremental Load** property to **Specific Date and Time** for an ingestion task in the task wizard, the inventory list is used to determine the start point in the logs when the database ingestion job runs for the first time or restarts. The log inventory list provides Mass Ingestion Databases with a starting RBA or LSN and an ending RBA or LRSN, along with the starting and ending timestamps of all active and archive logs for the Db2 subsystem. Mass Ingestion Databases uses this information to select the appropriate log archive to search for the requested start point. If the **Initial Start Point for Incremental Load** property is set to any option other than **Specific Date and Time**, you do not need to install the USERJCL library.

Contents of the LOGINV member:

```
//<USERID>I JOB  'LOG INVENTORY',MSGLEVEL=(1,1),MSGCLASS=X,
//      NOTIFY=&SYSUID,CLASS=A,REGION=0M
//* -----------------------------------------------------
//*
//*  PLEASE CHANGE DSN? TO THE DB2 SUBSYSTEM OF YOUR CHOICE.
//*
//*  THIS JCL MEMBER CAN BE USED TO RUN A LOG INVENTORY
//*  LIST. DBMI WILL REQUEST THIS IF A DBMI JOB IS TRYING TO
//*  RETART USING TIME. THE LOG INVERNTORY GIVES DBMI THE
//*  ABILITY TO CORRELATE RBA/LSRN TO ACTUAL LOG RESTART
//*  POSITIONS. DBMI PARSES THE SYSPRINT OUTPUT TO GET THE
//*  REQUIRED INFORMATION.
//*
//*
//*  SUBSTITUTION TAGS
//*  _________________
//*
//*  SUBSTITUTION TAGS ARE USED TO INSERT DATA INTO THE JOB
//*  BY DBMI BEFORE IT IS SUBMITTED. YOU MAY COPY THIS JCL
//*  INTO ANOTHER MEMBER AND MODIFY ITS CONTENTS. SUBSTITUTION
//*  TAGS MAY ALSO BE REMOVED AND HARD CODED FOR INDIVIDUAL
//*  JOB NEEDS.
//*
//*
//*  <USERID>     WILL BE REPLACED WITH THE USER ID.
//*
//* -----------------------------------------------------
//LOGINV EXEC PGM=DSNJU004
//STEPLIB DD DISP=SHR,DSN=DSN?10.SDSNLOAD
//        DD DISP=SHR,DSN=DSN?10.DSNC.RUNLIB.LOAD
//SYSUT1   DD DISP=SHR,DSN=DSN?.BSDS01
//SYSPRINT DD SYSOUT=*
//SYSSUMRY DD SYSOUT=*
//
```

Note: In this JCL, Mass Ingestion Databases replaces <USERID> with the user specified in the Db2 for z/OS Database Ingestion connection properties

To submit USERJCL member jobs and retrieve output from submitted jobs, Mass Ingestion Databases executes Db2-supplied stored procedures in batch as required. Make sure that the Mass Ingestion Databases user has the required privileges to execute these stored procedures. For more information about Db2-supplied procedures, see https://www.ibm.com/docs/en/db2-for-zos/12?topic=sql-procedures-that-are-supplied-db2.

A USERJCL member is processed in the following manner:

1. For the LOGINV member, if the **Initial Start Point for Incremental Load** advanced property is set to **Specific Date and Time** on the **Source** page of the database ingestion task wizard, Mass Ingestion Databases checks whether the following custom properties are specified for the source:
   - **pwx.cdcreader.ZOS.Db2JobsDSN**, which specifies the name of the installed USERJCL PDS or PDSEd
   - **pwx.cdcreader.ZOS.Db2JobLLOGINVMember**, which specifies the LOGINV member name that will be used for a database ingestion job.
2. Mass Ingestion Databases reads the LOGINV member by using the Db2-supplied stored procedure ADMIN_DS_BROWSE.
3. Mass Ingestion Databases substitutes any tags required for the job execution.
4. Mass Ingestion Databases submits the job to the Db2 for z/OS database by using the Db2-supplied stored procedure ADMIN_SUBMIT_JOB.
5. Mass Ingestion Databases queries the status of the submitted jobs by using the Db2-supplied stored procedure ADMIN_JOB_QUERY.
6. After the job completes, Mass Ingestion Databases obtains the job output by using the Db2-supplied stored procedure ADMIN_JOB_FETCH.

Db2 for z/OS privileges

To deploy and run a database ingestion task that has a Db2 for z/OS source, the source connection must specify a Mass Ingestion Databases user who has the privileges required for the ingestion load type.

Privileges for initial load processing

- For initial load jobs, grant the following privilege to the user specified in the Db2 for z/OS database ingestion connection properties:

  ```
  SELECT on schema.table  TO user
  ```

  Where *schema*.*table* identifies a source table.
- To run queries against Db2 catalog tables during source unload processing, grant the SELECT privilege on the following catalog tables:

  - SYSIBM.SYSCOLUMNS
  - SYSIBM.SYSINDEXES
  - SYSIBM.SYSKEYS
  - SYSIBM.SYSTABLEPART
  - SYSIBM.SYSTABLES
  - SYSIBM.SYSTABLESPACE

Privileges for incremental load processing

For incremental load jobs, grant the following privileges.

Note: If you do not have the authority required to issue these grants, ask a Db2 administrator who has SYSDBA authority or a higher authority level to issue them.

- To generate the CDC script that enables the database CDC option from the user interface, issue the following grant to the interface user:

  ```
  GRANT ALTER TABLE schema.table  DATA CAPTURE CHANGES TO user   <--for each source table
   COMMIT;
  ```

- For incremental change data processing, grant the following privileges to the user specified in the Db2 for zOS Database Ingestion connection properties:
  - To make the change data in the Db2 logs available to the stored procedure:

    ```
    GRANT ALTER TABLE schema.table  DATA CAPTURE CHANGES TO user   <--for each source table
    COMMIT;
    ```

  - To enable the user to obtain required information from the Db2 Instrumentation Facility Interface (IFI):

    ```
    GRANT MONITOR2 TO user;
    COMMIT;
    ```

  - To grant authorities on the global temporary table to the user:

    ```
    GRANT READ ON !SCHEMA!.!STRPRC!_RS_TBL to user;
    GRANT DELETE ON !SCHEMA!.!STRPRC!_RS_TBL to user;
    GRANT EXECUTE !SCHEMA!.!STRPRC! to user;
    COMMIT;
    ```

    Where !SCHEMA! and !STRPRC! are variables in the job JCL, which represent the stored procedure schema name and procedure name respectively.

    The first two privileges allow the user to read and delete the contents of the Global Temporary Table to which the stored procedure is writing the log data. The third privilege allows the user to run the stored procedure.

  - To enable the stored procedure to bind its Db2 plan, grant the following privileges to PUBLIC:

    ```
    GRANT BIND, EXECUTE ON PLAN !STRPRC! TO PUBLIC;
    COMMIT;
    ```

  - To run queries against Db2 catalog tables during source unload processing, grant the SELECT privilege on the following catalog tables:
    - SYSIBM.SYSCOLUMNS
    - SYSIBM.SYSDUMMY1
    - SYSIBM.SYSINDEXES
    - SYSIBM.SYSKEYS
    - SYSIBM.SYSTABLEPART
    - SYSIBM.SYSTABLES
    - SYSIBM.SYSTABLESPACE

Permissions Required for Executing the Stored Procedure on z/OS

Ensure that the following Db2 permissions are granted before you run the stored procedure JCL job:

- Ensure that the user who executes the stored procedure job has SYSADM authority, or ask the Db2 for z/OS DBA to run it.
- For the stored procedure to run, you must grant the following Db2 permissions to the procedure schema name specified in the #STPINST JCL file:
  - SELECT authority on Db2 catalog tables:

    ```
    GRANT SELECT ON SYSIBM.* TO schema;
    ```

  - EXECUTE authority on the package name specified in the JCL.

    ```
    GRANT EXECUTE ON PACKAGE package_name TO schema;
    ```

Additionally, grant INSERT and DELETE authority on the schema and stored procedure name that are specified in the JCL for the global temporary table:

```
GRANT INSERT, DELETE ON schema.stored_procedure_name_RS_TBL TO user
```

Snowflake targets

Target preparation

Complete the following steps as the ACCOUNTADMIN user.

1. Create a Mass Ingestion user. Use one of the following SQL statements:

```
create user INFACMI_User password  'Xxxx@xxx';
```

or

```
replace user INFACMI_User password  'Xxxx@xxx';
```

2. Create a new role and grant the role to the Mass Ingestion user. Use the following SQL statements:

```
create role INFA_CMI_Role;
grant role INFA_CMI_Role to user INFACMI_User;
```

3. Grant usage on the Snowflake virtual warehouse to the new role. Use the following SQL statement:

```
grant usage on warehouse CMIWH to role INFA_CMI_Role;
```

4. Grant usage on the Snowflake database to the new role. Use the following SQL statement:

```
grant usage, CREATE SCHEMA on database CMIDB to role INFA_CMI_Role;
```

5. Set the default role for the newly created user. Use the following SQL statement:

```
alter user INFACMI_User set default_role=INFA_CMI_Role;
```

Also, as the INFACMI_User, create a new schema:

```
create schema CMISchema;
```

Note: If the user's default role is used for ingestion tasks and does not have the required privileges, the following error will be issued at runtime:

```
SQL compilation error: Object does not exist, or operation cannot be performed.
```

Usage considerations

- Before writing data to Snowflake target tables, database ingestion jobs first write the data to an internal stage with the name you specified in the associated database ingestion task.
- When you define a connection for a Snowflake target, you must set the **Additional JDBC URL Parameters** field to `database=target_database_name`. Otherwise, when you try to define the target in the database ingestion task wizard, an error message reports that the list of schemas cannot be retrieved.
- When you define a connection for a Snowflake target using the **KeyPair** option as the authentication method and you generate the private key with OpenSSL 3.x.x version, use `PBE-SHA1-2DES` or `PBE-SHA1-3DES` cipher while generating the private key. Run one of the following commands:

```
openssl genrsa 2048 | openssl pkcs8 -topk8 -inform PEM -v1 PBE-SHA1-3DES -out rsa_key.p8
```

or:

```
openssl genrsa 2048 | openssl pkcs8 -topk8 -inform PEM -v1 PBE-SHA1-2DES -out rsa_key.p8
```

If you use a generic command without the `PBE-SHA1-2DES` or `PBE-SHA1-3DES` cipher, an error message about an invalid or unsupported private key might occur while fetching the target schema in the target definition step of the database ingestion task wizard.
The error message does not occur if you use OpenSSL 1.1.1 to generate the private key.

- Database ingestion incremental load jobs and combined initial and incremental load jobs generate a recovery table named INFORMATICA_CDC_RECOVERY on the target to store internal service information that prevents jobs restarted after a failure from propagating previously processed data again. This recovery table is generated in the same schema as the target tables.
- For Snowflake targets, you cannot alter the scale of NUMBER fields or change the data type of an existing field to a different data type because Snowflake does not support these actions.

Configuring connections

Configure a source connection and a target connection, if not already defined, from the **Connections** page in Administrator. Mass Ingestion Databases uses the Db2 for zOS Database Ingestion connector and Snowflake Data Cloud connector.

1. In Administrator, select **Connections**.
2. On the **Connections** page, click **New Connection**.
3. Configure the following connection details:

Property	Description
Connection Name	The name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Maximum length is 100 characters. Connection names are not case sensitive.
Description	A description of the connection.
Type	The type of connection, such as Snowflake Data Cloud.

   After you select the connection type, additional properties that are specific to that type appear.

4. Configure the Db2 for z/OS connection properties and the Snowflake connection properties.
5. To test each connection, click **Test Connection**.

   The results of the test are displayed at the top of the page.

   If a connection fails, contact the database administrator, or recheck your settings and verify that the selected runtime environment has the status of Up and Running.
6. Click **Save** to save the connection.

Db2 for zOS Database Ingestion connection properties

When you define a Db2 for zOS Database Ingestion connection, you must configure connection properties. You can use this connection type in database ingestion tasks, which you configure in the Mass Ingestion service.

The following table describes the connection properties:

Property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Type	The type of connection. Ensure that the type is Db2 for zOS Database Ingestion.
Runtime Environment	The name of the runtime environment where you want to run database ingestion tasks. You define runtime environments in Administrator.

Property	Description
User Name	The user name to use for connecting to the Db2 for zOS instance.
Password	The password to use for connecting to the Db2 for zOS instance.
Host	The name of the machine that hosts the database server.
Port	The network port number used to connect to the database server.
Location Name	The name of the Db2 for zOS location that you want to access. For DB2 for z/OS, your system administrator can determine the name of your DB2 location using the command DISPLAY DDF.
Code Page for Bit Data	The code page that Mass Ingestion Databases uses to read character data that is stored as bit data. This value must be a canonical name for the java.io API and java.lang API. For more information, see the supported encodings in the Oracle Java documentation. Specify this property if you have FOR BIT DATA source columns.
CDC Stored Procedure Schema	For incremental change data capture processing, the name of the schema for the z/OS stored procedure that is required to collect change data from the Db2 log. This value is specified in the #STPINST data set that you customized when setting up the stored procedure on z/OS. No default value is provided.
CDC Stored Procedure Name	For incremental change data capture processing, the name of the z/OS stored procedure that is required to collect change data from the Db2 log. This value is specified in the #STPINST data set that you customized when setting up the stored procedure on z/OS. The default value is INFALOG.
Advanced Connection Properties	Advanced properties for the Progress DataDirect JDBC DB2 driver, which is used to connect to the Db2 for z/OS source. If you specify more than one *property=value* entry, separate them with a semicolon (;). The driver properties that you can enter in this field are described in the Progress DataDirect documentation at https://docs.progress.com/bundle/datadirect-connect-jdbc-51/page/Connection-Properties_10.html. For example, you can set the ConnectionRetryCount property to control the number of times the driver retries attempts to connect to the primary database server.

Snowflake Data Cloud connection properties

When you set up a Snowflake Data Cloud connection, configure the connection properties.

You can use the following authentication methods to connect to Snowflake:

- Standard. Uses Snowflake account user name and password credentials to connect to Snowflake.

  Note: For application ingestion tasks, you can use only the Standard authentication method.
- Authorization Code. Uses the OAuth 2.0 protocol with Authorization Code grant type to connect to Snowflake. Authorization Code allows authorized access to Snowflake without sharing or storing your login credentials.
- KeyPair. Uses the private key file and private key file password, along with the existing Snowflake account user name to connect to Snowflake.
- Client Credentials. Uses the OAuth 2.0 protocol with the Client Credentials grant type to connect to Snowflake. Doesn't apply to Mass Ingestion application and database ingestion tasks.

You create a Snowflake Data Cloud connection on the Connections page. You can then use the connection when you read from or write data to Snowflake.

Standard authentication

When you set up a Snowflake Data Cloud connection, configure the connection properties.

The following table describes the Snowflake Data Cloud connection properties for the Standard authentication mode:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	The Snowflake Data Cloud connection type.
Runtime Environment	The name of the runtime environment where you want to run the tasks. You cannot run application ingestion tasks and database ingestion tasks on a Hosted Agent or serverless runtime environment.
Authentication	The authentication method that the connector must use to log in to Snowflake. Select **Standard**. Default is **Standard**.
Username	The user name to connect to the Snowflake account.
Password	The password to connect to the Snowflake account.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws **Note:** Ensure that the account name doesn't contain underscores. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.
Role	The Snowflake role assigned to the user.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	Enter one or more JDBC connection parameters in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, pass the database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	Important: Ensure that there is no space before and after the equal sign (=) when you add the parameters.

OAuth 2.0 authorization code authentication

The following table describes the Snowflake Data Cloud connection properties for an OAuth 2.0 - AuthorizationCode type connection:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	The Snowflake Data Cloud connection type.
Runtime Environment	The name of the runtime environment where you want to run the tasks. You cannot run application ingestion tasks and database ingestion tasks on a Hosted Agent or serverless runtime environment.
Authentication	The authentication method that Snowflake Data Cloud Connector must use to log in to Snowflake. Select **AuthorizationCode**.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws **Note:** Ensure that the account name doesn't contain underscores. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	Enter one or more JDBC connection parameters in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, pass the database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	Important: Ensure that there is no space before and after the equal sign (=) when you add the parameters.

Authorization URL	The Snowflake server endpoint that is used to authorize the user request. The authorization URL is https://<account name>.snowflakecomputing.com/oauth/authorize, where <account name> specifies the full name of your account provided by Snowflake. For example, https://<abc>.snowflakecomputing.com/oauth/authorize **Note:** If the account name contains underscores, use the alias name. You can also use the Authorization Code grant type that supports the authorization server in a Virtual Private Cloud network.

Property	Description
Access Token URL	The Snowflake access token endpoint that is used to exchange the authorization code for an access token. The access token URL is https://<account name>.snowflakecomputing.com/oauth/token-request, where <account name> specifies the full name of your account provided by Snowflake. For example, https://<abc>.snowflakecomputing.com/oauth/token-request **Note:** If the account name contains underscores, use the alias name.
Client ID	Client ID of your application generated when you create a security integration of type OAuth in Snowflake. For more information, see the Snowflake documentation. Not used by Mass Ingestion application and database ingestion tasks.
Client Secret	Client secret generated for the client ID. Not used by Mass Ingestion application and database ingestion tasks.
Scope	Determines the access control if the API endpoint has defined custom scopes. Enter space-separated scope attributes. For example, specify `session:role:CQA_GCP` as the scope to override the value of the default user role. The value must be one of the roles assigned in Security Integration. Not used by Mass Ingestion application and database ingestion tasks.
Access Token Parameters	Additional parameters to use with the access token URL. Define the parameters in the JSON format. For example, define the following parameters: `[{"Name":"code_verifier","Value":"5PMddu6Zcg6Tc4sbg"}]` Not used by Mass Ingestion application and database ingestion tasks.

Authorization Code Parameters	Additional parameters to use with the authorization token URL.
	Define the parameters in the JSON format.
	For example, define the following parameters:
	`[{"Name":"code_challenge","Value":"Ikr-vv52th0UeVRi4"},`
	`{"Name":"code_challenge_method","Value":"S256"}]`
	Not used by Mass Ingestion application and database ingestion tasks.

Access Token	The access token value. Enter the populated access token value, or click **Generate Token** to populate the access token value.
Generate Token	Generates the access token and refresh token based on the OAuth attributes you specified.
Refresh Token	The refresh token value. Enter the populated refresh token value, or click **Generate Token** to populate the refresh token value. If the access token is not valid or expires, the agent fetches a new access token with the help of the refresh token. **Note:** If the refresh token expires, provide a valid refresh token or regenerate a new refresh token by clicking **Generate Token**. Not used by Mass Ingestion application and database ingestion tasks.

Key pair authentication

The following table describes the Snowflake Data Cloud connection properties for the KeyPair authentication type connection:

Connection property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	The Snowflake Data Cloud connection type.
Runtime Environment	The name of the runtime environment where you want to run the tasks. You cannot run application ingestion tasks and database ingestion tasks on a Hosted Agent or serverless runtime environment.
Authentication	The authentication method to log in to Snowflake. Select **KeyPair**.
Username	The user name to connect to the Snowflake account.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	Enter one or more JDBC connection parameters in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, pass the database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	Important: Ensure that there is no space before and after the equal sign (=) when you add the parameters.

Private Key File	Path to the private key file, including the private key file name, that the Secure Agent uses to access Snowflake. For example, specify the following path and key file name: - On Windows: `C:\Users\path_to_key_file\rsa_key.p8` - On Linux: `/export/home/user/path_to_key_file/rsa_key.p8` **Note:** Verify that the keystore is FIPS-certified.
Private Key Password	Password for the private key file.

Configuring a database ingestion task

In Mass Ingestion, use the database ingestion task wizard to configure a database ingestion task.

On the wizard pages, complete the following configuration tasks:

1. Define basic task information, such as the task name, project location, runtime environment, and load type.
2. Configure the source.
3. Configure the target.
4. Configure runtime options.

Click **Next** or **Back** to navigate from one page to another. At any point, you can click **Save** to save the information that you have entered so far.

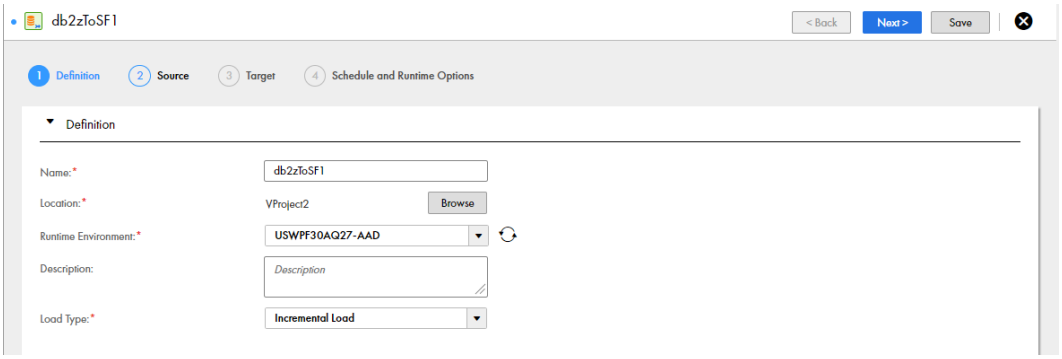
After you complete all wizard pages, save the information and then click **Deploy** to make the task available as an executable job to the Secure Agent.

Step 1: Defining basic task information

To begin defining a database ingestion task, you must first enter some basic information about the task, such as a task name, project or project folder location, and load operation type.

1. In Mass Ingestion, click **New** > **Database Ingestion Task**.

   The **Definition** page of the Mass Ingestion Databases Task wizard appears.



2. Configure the following properties:

Property	Description
Name	A name for the database ingestion task. Task names can contain Latin alphanumeric characters, spaces, periods (.), commas (,), underscores (_), plus signs (+), and hyphens (-). Task names cannot include other special characters. Task names are not case sensitive. Maximum length is 50 characters. **Note:** If you include spaces in the database ingestion task name, after you deploy the task, the spaces do not appear in the corresponding job name.
Location	The project or project\folder that will contain the task definition. The default is the currently selected project or project subfolder in Explore. If a project or project subfolder is not selected, the default is the **Default** project.

Property	Description
Runtime Environment	The runtime environment in which you want to run the task. The runtime environment must be a Secure Agent group that consists of one or more Secure Agents. A Secure Agent is a lightweight program that runs tasks and enables secure communication. You cannot use the Hosted Agent or a serverless runtime environment. **Tip:** Click the **Refresh** icon to refresh the list of runtime environments.
Description	An optional description for the task. Maximum length is 4,000 characters.
Load Type	The type of load operation that the database ingestion task performs. In this scenario, select **Incremental Load** to propagate source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run.

3. Click **Next**.

Step 2: Configuring the Db2 for z/OS source

Configure the Db2 for z/OS source on the **Source** page of the database ingestion task wizard.

1. In the **Connection** list, select the connection for the Db2 for z/OS source that you configured and tested in Administrator for a runtime environment that your organization uses. The connection type appears in parentheses after the connection name.

   The list includes only the connection types that are valid for the load type selected on the **Definition** page. No connections are listed if you did not select a load type.

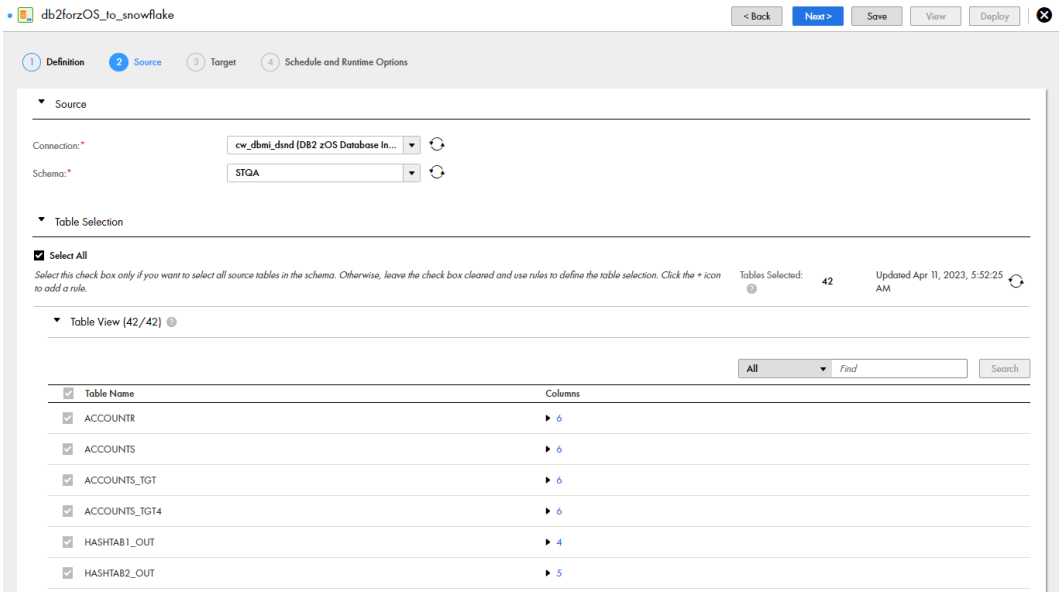
   Note: After you deploy the ingestion task, you cannot change the connection without first undeploying the associated ingestion job. After you change the connection, you must deploy the task again.
2. In the **Schema** list, select the source schema that includes the source tables.

   The list includes only the schemas that are available in the database accessed with the specified source connection.
3. Under **Table Selection**, select **Select All** only if you want to select all tables and columns in the schema for data replication. You cannot edit the selection in subsequent fields.

   The **Tables Selected** field shows the count of all selected tables. The interface fetches information about all of the tables.

   Note: Fetching information for all of the tables might take a long time.

   The following image shows the **Source** page when **Select All** is selected:



Alternatively, you can use rules to define a subset of source tables to replicate.

4. To use rules to select the source tables, make sure that the **Select All** check box is cleared and then add rules.

   When rule-based selection is used, you can refine the set of selected tables by table under **Table View** and also set an option for trimming spaces in character data.

   Note: The default "Include *" rule selects all tables in the specified schema. To see how many tables are selected by this rule, click the Refresh icon to display the table count in **Total Tables Selected** and click **Apply Rules** to see the table count in **Table View**.

   To add a rule:

   a. Under **Rules**, click the Add Rule (+) icon above the first table. A row is added to the table.

   b. In the **Table Rule** column, select **Include** or **Exclude** to create an inclusion or exclusion rule, respectively.

   c. In the **Condition** column, enter a table name or a table-name mask that includes one or more wildcards to identify the source tables to include in or exclude from table selection. Use the following guidelines:

      - A mask can contain one or both of the following wildcards: an asterisk (*) wildcard to represent one or more characters and a question mark (?) wildcard to represent a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
      - The task wizard is case sensitive. Enter the table names or masks in the case with which the tables were defined.
      - Do not include delimiters such as quotation marks or brackets, even if the source database uses them.
      - If a table name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

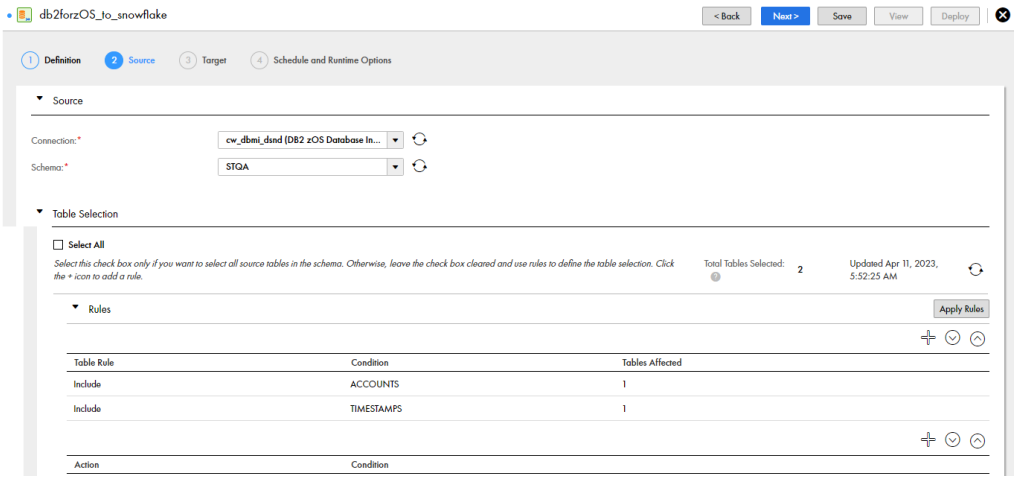
   d. Define additional rules as needed.

      The rules will be processed in the order in which they're listed, from top to bottom. Use the arrow icons to change the order.

e. When finished, click **Apply Rules**.

The **Total Tables Selected** and **Tables Affected** counts are updated.

The following image shows multiple rules defined on the **Source** page:



After you apply rules, if you add, delete, or change rules, you must click **Apply Rules** again. Click the Refresh icon to update the table counts. If you delete all rules without clicking **Apply Rules**, a validation error will occur at deployment, even if the **Table View** list still lists tables. If you switch to **Select All**, the rules are no longer in effect and disappear.

5. To perform trim actions on character columns in the source tables selected based on rules, create column action rules.

   To create a column action rule:

   a. Click the Add Rule (+) icon above the second Action table under **Rules**.

   b. In the **Action** column, select one of the following options:

      - **LTRIM**. Trims spaces to the left of character column values.
      - **RTRIM**. Trims spaces to the right of character column values.
      - **TRIM**. Trims spaces to the left of and to the right of character column values.

   c. In the **Condition** column, enter a column name or a column name mask that includes one or more asterisk (*) or question mark (?) wildcards. The value is matched against columns in the selected source tables to identify the columns to which the action applies.

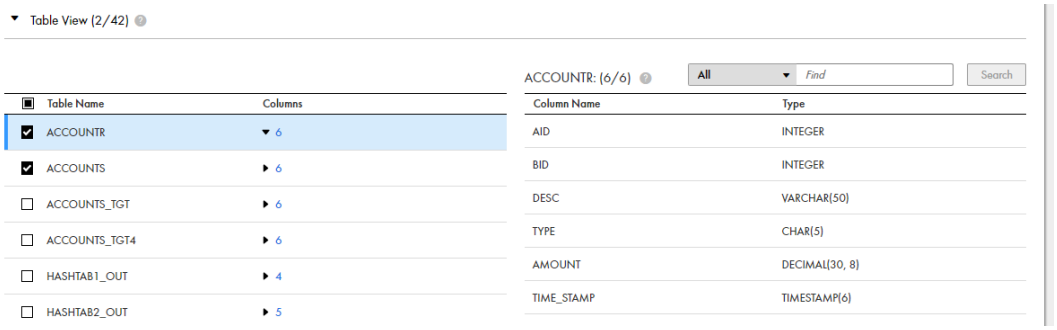
   Note: You can define multiple rules for different action types or for the same action type with different conditions. The rules are processed in the order in which they're listed, from top to bottom. Use the arrow icons to change the order.

6. Under **Table View**, view or edit the set of selected source tables and columns.

   If you selected **Select All**, the lists of tables and columns are view only.

   If you applied rules, you can refine the set of selected tables by clicking the check box next to individual tables. Deselect any tables that you do not want to replicate, or select additional items to replicate. Click the Refresh icon to update the selected tables count.

   The following image shows selected tables:



Notes:

- To search for tables or columns or both, in the drop-down list above the columns list, select **Table Name**, **Columns**, or **All** and then enter a search string in the *Find* box and click **Search**. You can include a single asterisk (*) wildcard at the beginning or end of the string.
- The first time you change a check box setting for a table or column, the rules are no longer in effect. The selections under **Table View** take precedence. However, if you click the Add Rule (+) icon again, any tables that you deselected or selected individually are reflected as new rules in the Rules list and the rules once again take precedence. If you want to return to the **Table View** list, click **Apply Rules** again.
- In the **Table View** section, the length of character columns is shown in bytes. Since the number of bytes per character differs depending on the character set encoding used by the database, for some sources, the actual column length in number of characters might be different than the number of bytes shown in the **Table View**.

7. If one or more of the selected source tables are not enabled for change data capture, you can generate a script for enabling CDC and then run or download the script.

   a. In the **CDC Script** field, select **Enable CDC for all columns** to enable CDC for all columns in the selected source tables. This option is the only available option for Db2 for z/OS sources.

      The script sets DATA CAPTURE CHANGES for source tables and certain Db2 catalog tables needed for CDC. After DATA CAPTURE CHANGES is set for one job, all other jobs recognize that the attribute is enabled in Db2 for the catalog tables needed because the Db2 catalog tables are a set of tables shared by all users of Db2.

   b. To run the script, click **Execute**.

      If you do not have a database role or privilege that allows you to run the script, click the Download icon to download the script. The script file name has the following format: `cdc_script_taskname_number.txt`. Then ask your database administrator to run the script.

      Make sure the script runs before you run the database ingestion task.

   Note: If you change to the **CDC Script** option later and run the script again, the script first drops CDC for the original set of columns and then enables CDC for the current set of columns.

8. To create and download a list of the source tables that match the table selection criteria, perform the following substeps:

   a. If you used rule-based table selection, in the **List Tables by Rule Type** list, select the type of selection rules that you want to use. Options are:

      - **Include Rules Only**
      - **Exclude Rules Only**
      - **Include And Exclude Rules**

b. To include columns in the list, regardless of which table selection method you used, select the **Include Columns** check box.

c. Click the Download icon.

A downloaded list that includes columns has the following format:

```
status,schema_name,table_name,object_type,column_name,comment
```

The following table describes the information that is displayed in the downloaded list:

Field	Description
status	Indicates whether Mass Ingestion Databases excludes the source table or column from processing because it has an unsupported type. Valid values are: - **E**. The object is excluded from processing by an Exclude rule. - **I**. The object is included in processing. - **X**. The object is excluded from processing because it is an unsupported type of object. For example, unsupported types of objects include columns with unsupported data types and tables that include only unsupported columns. The comment field provides detail on unsupported types.
schema_name	Specifies the name of the source schema.
table_name	Specifies the name of the source table.
object_type	Specifies the type of the source object. Valid values are: - **C**. Column. - **T**. Table.
column_name	Specifies the name of the source column. This information appears only if you selected the **Columns** check box.
comment	Specifies the reason why a source object of an unsupported type is excluded from processing even though it matches the selection rules.

9. Under **Advanced**, set the advanced properties that are available for incremental load jobs with Db2 for z/OS sources:

Property	Description
Enable Persistent Storage	Select this check box to enable persistent storage of transaction data in a disk buffer so that the data can be consumed continually, even when the writing of data to the target is slow or delayed. Benefits of using persistent storage are faster consumption of the source transaction logs, less reliance on log archives or backups, and the ability to still access the data persisted in disk storage after restarting a database ingestion job.
Initial Start Point for Incremental Load	Set this field if you want to customize the position in the source logs from which the database ingestion job starts reading change records the first time it runs. Options are: - **Earliest Available**. The earliest available record in the Db2 for z/OS transaction log. - **Latest Available**. The latest available position in the database log. - **Position**. A valid LRSN that is used to determine a position in the change stream from which to start retrieving change records. The LRSN must be equal to or less than the current LRSN. An invalid value will cause the job to fail. The default value is 0, which causes processing to start from the latest available position. - **Specific Date and Time**. A date and time, in the format MM/DD/YYYY hh:mm AM|PM, that Mass Ingestion Databases uses to determine the position in the change stream from which to start retrieving change records. Mass Ingestion Databases retrieves only the changes that were started after this date and time. If you enter a date and time earlier than the earliest date and time in the available archived logs, the job will fail. The default is **Latest Available**. **Note:** The **Initial Start Point for Incremental Load** option pertains only to the initial run of a job. Thereafter, if you resume a stopped or aborted job, the job begins propagating source data from where it last left off.

10. Under **Custom Properties**, you can specify custom properties at the direction of Informatica Global Customer Support to meet your special requirements. In this scenario, skip this step.
11. Click **Next**.

Step 3: Configuring the Snowflake target

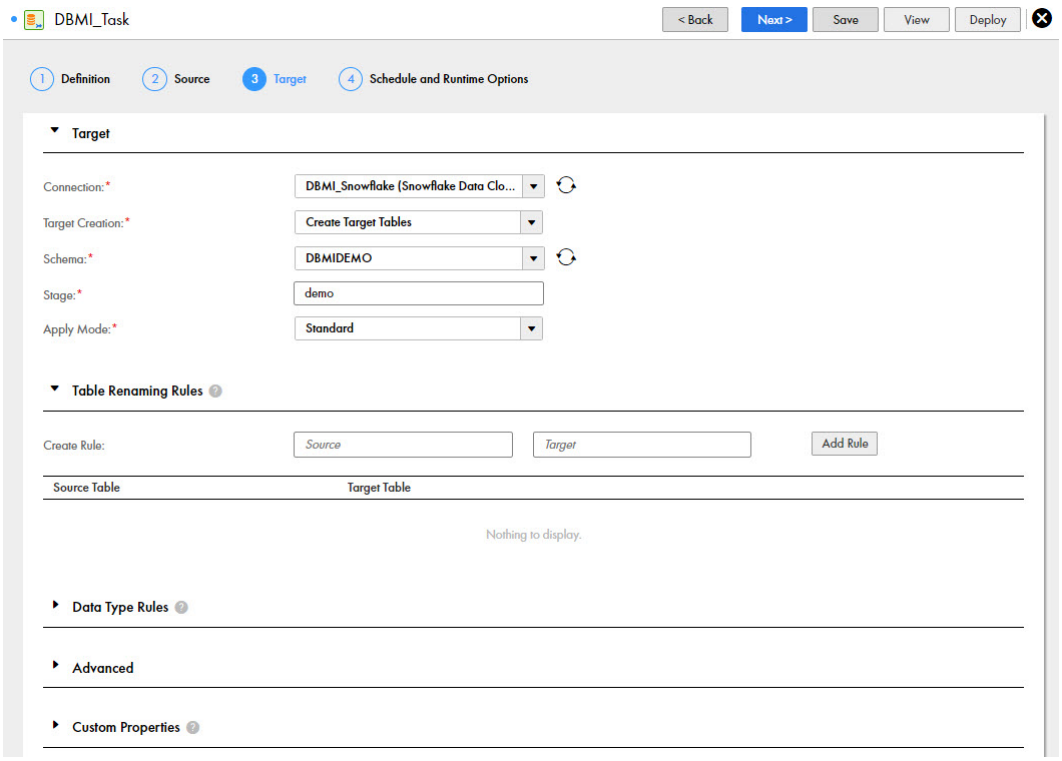
Configure the target on the **Target** page of the database ingestion task wizard.

1. In the **Connection** list, select the Snowflake connection that you previously defined and tested in Administrator for the runtime environment. The connection type appears in parentheses after the connection name.

   The list includes only the connection types that are valid for the load type selected on the **Definition** page. No connections are listed if you did not select a load type.

   If you change the load type and the selected connection is no longer valid, a warning message is issued and the **Connection** field is cleared. You must select another connection that is valid for the updated load type.

   Note: After you deploy the ingestion task, you cannot change the connection without first undeploying the associated ingestion job. You must then deploy the task again.
2. In the **Target** section, configure the properties that pertain to a Snowflake target.



The following table describes these properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source tables. **Note:** After the target table is created, Mass Ingestion Databases intelligently handles the target tables on subsequent job runs. Mass Ingestion Databases might truncate or re-create the target tables depending on the specific circumstances.
Schema	Select the target schema in which Mass Ingestion Databases creates the target tables.

Property	Description
Stage	The name of internal staging area that holds the data read from the source before the data is written to the target tables. This name must not include spaces. If the staging area does not exist, it will be automatically created.
Apply Mode	Indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are: - **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target. - **Soft Deletes**. Apply source delete operations to the target as soft deletes. A soft delete marks the deleted row as deleted without actually removing it from the database. For example, a delete on the source results in a change record on the target with "D" displayed in the INFA_OPERATION_TYPE column. If an update followed by a delete occurs on the source, two records are written to the target both with "D" displayed in the INFA_OPERATION_TYPE column. Consider using soft deletes if you have a long-running business process that needs the soft-deleted data to finish processing, to restore data after an accidental delete operation, or to track deleted values for audit purposes. **Note:** If you use **Soft Deletes** mode, you must not perform an update on the primary key in a source table. Otherwise, data corruption can occur on the target. - **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, time, owner, transaction ID, generated ascending sequence number, and before image. Consider using Audit apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes. The default value is **Standard**. ***In this scenario, accept the default value of **Standard**.***

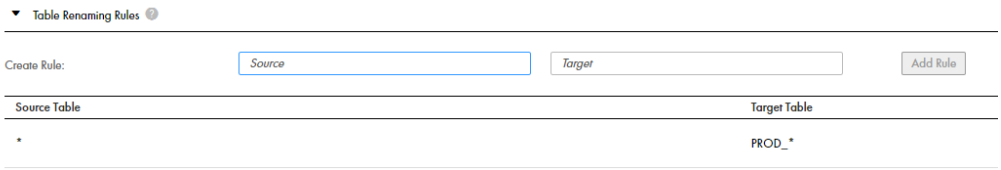
3. If you want to rename the target objects that are associated with the selected source tables, define table renaming rules. For example, you might want to add a prefix or suffix. To create a rule for renaming tables:
   a. Under **Table Renaming Rules**, in the **Create Rule** fields, enter a source table name or a table name mask that includes one or more wildcards. Then enter the corresponding target table name or table name mask.

      Notes:

      - For the source, you can enter only the asterisk (*) wildcard to select all source tables that match the table selection criteria on the **Source** page. Or you can enter a specific source table name or a table-name mask that includes one or more of the following wildcards: an asterisk (*) to represent one or more characters or a question mark (?) to represent a single character.
      - To use a table-name mask with the wildcard character for the target, you must also use the wildcard character in the source. If you use a specific source table name with a target table mask that includes the wildcard character, the task deployment will fail.
      - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).

b. Click **Add Rule**

The rule appears in the rules list.



You can define multiple table rules. The order of the rules does not matter with regard to how they are processed unless a table matches multiple rules. In this case, the last matching rule determines the name of the table.

4. If you want to override the default mappings of source data types to target data types, define data type rules.
5. Under **Advanced**, you can enter the following advanced target properties which will occur in target tables:

Field	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. The default value is INFA_.
Suprpipe	Snowpipe Streaming API allows you to quickly stream rows of data directly to Snowflake Data Cloud target tables with low latency instead of first writing the data to stage files. By default, this check box is selected. ***In this scenario, **deselect** the check box.***
Merge Frequency	When **Superpipe** is selected, you can optionally set the frequency, in seconds, at which change data rows are merged and applied to the Snowflake target tables.

6. Under **Custom Properties**, you can specify custom properties at the direction of Informatica Global Customer Support to meet your special requirements. In this scenario, skip this step.
7. Click **Next**.

Step 4: Configuring runtime options

On the **Schedule and Runtime Options** page of the database ingestion task wizard, you can configure some runtime options.

1. Under **Schema Drift Options**, specify the schema drift option to use for each of the supported types of DDL operations.

   The types of supported DDL operations are:

   - Add Column
   - Modify Column

- Drop Column
- Rename Column

The following table describes the schema drift options that you can set for a DDL operation type:

Option	Description
Ignore	Do not replicate DDL changes that occur on the source database to the target. For Snowflake targets, this option is the default option for the Drop Column and Rename Column operation types
Replicate	Replicate the DDL operation to the target. For Snowflake targets, this option is the default option for the Add Column and Modify Column operation types. **Restrictions:** - If you try to replicate a type of schema change that is not supported on the target, database ingestion jobs associated with the task will end with an error. - Add Column operations that add a primary-key column are not supported and can cause unpredictable results. - Modify Column operations that change the NULL or NOT NULL constraint for a column are not replicated to the target by design because changing the nullibility of a target column can cause problems when subsequent changes are applied. - DDL operations are replicated to the target only after some data changes have been captured.
Stop Job	Stop the entire database ingestion job.
Stop Table	Stop processing the source table on which the DDL change occurred. When one or more of the tables are excluded from replication because of the **Stop Table** schema drift option, the job state changes to **Running with Warning**. **Important:** The database ingestion job cannot retrieve the data changes that occurred on the source table after the job stopped processing it. Consequently, data loss might occur on the target. To avoid data loss, you will need to resynchronize the source and target objects that the job stopped processing. Use the **Resume With Options** > **Resync** option.

2. Under **Advanced**, optionally edit the **Number of Rows in Output File** value to specify the maximum number of rows that the database ingestion task writes to an output data file for the Snowflake target.

   Change data is flushed to the target either when this number of rows is reached or when the flush latency period expires and the job is *not* in the middle of processing a transaction. The flush latency period is the time that the job waits for more change data before flushing data to the target. The latency period is internally set to 10 seconds and cannot be changed.

   Valid values are 1 through 100000000. The default value is 100000 rows.

   Note: For Snowflake targets, the data is first stored in an internal stage area before being written to the target tables.

3. Under **Schedule**, the scheduling options are unavailable for incremental load tasks or combined initial and incremental load. Skip this field.
4. Under **Custom Properties**, you can specify custom properties that Informatica provides to meet your special requirements. Specify these properties only at the direction of Informatica Global Customer Support.
5. Click **Save**.

Deploying the database ingestion task

After you define a database ingestion task and save it, deploy the task to create an executable job instance on the on-premises system that contains the Secure Agent and the Database Ingestion agent service and DBMI packages. You must deploy the task before you can run the job. The deploy process also validates the task definition.

Before you deploy a task that has a Snowflake target, drop any existing target tables that do not match the structure of the source tables, for example, because of added or dropped source columns or altered column null constraints or data types. When you deploy the task, the target tables are generated based the latest source structure.

1. In the database ingestion task wizard, save the completed task definition.
2. To deploy a task, click **Deploy**.

After you deploy a task successfully, the associated job instance has the status of Deployed. You can run the job from either the **My Jobs** page in Mass Ingestion or from the **All Jobs** tab on the **Mass Ingestion** page in Operational Insights.

Deployment considerations:

- If you included spaces in the database ingestion task name, the spaces are omitted from the corresponding job name for the generated job instance.
- If the deployment process fails, the job status switches to Failed. You can then Undeploy the job. To diagnose the error, download the error log from the **My Jobs** page in Mass Ingestion or from the **All Jobs** tab on the **Mass Ingestion** page in Operational Insights. In the Actions menu for the job, click **Error Log**. After you resolve the problem, deploy the task again from the database ingestion task wizard.
- If you undeploy a job and then want to run a job for the associated ingestion task again, you must deploy the task again to create a new job instance. The new job instance name ends with an incremented number in the format *taskname-job_instance_number*. The job instance number is incremented each time you deploy the ingestion task by adding 1 to the maximum instance number across all ingestion jobs.
- If the Secure Agent is restarted while the task is deploying, the job status switches to Failed. Avoid restarting the Secure Agent while tasks are being deployed.
- If a task appears to be hung in the Deploying state, restart the Secure Agent. The associated job instance acquires the status of Failed. You can undeploy it and then deploy it again.

Running the deployed database ingestion job

You can run a database ingestion job after it has been deployed and is in a state other than Undeployed from either the **My Jobs** page in the Mass Ingestion service or from the **All Jobs** tab on the **Mass Ingestion** page in Operational Insights.

1. Navigate to the row for the job that you want to run.
2. In the **Actions** menu at the right end of the row, click **Run**.

   A subtask is started for each source table.

Monitoring database ingestion jobs

You can monitor the progress, performance, and status of ingestion jobs from the Mass Ingestion and Operational Insights services.

Depending on the service you use and type of ingestion job, you can view following monitoring information:

- On the **My Jobs** page in the Mass Ingestion service, monitor the ingestion jobs for the ingestion tasks that you deployed. You can view a list of your jobs that includes general job properties such as the task type, runtime environment, start time, duration, and current job status.

- On the **Mass Ingestion** page in Operational Insights service, monitor *all* types of ingestion jobs that any member of your organization deployed. You can view the following types of information:
  - Summary counts of ingestion jobs by task type and job status.
  - Recent jobs that require your attention because they failed or are running with errors or warnings.
  - A list of all ingestion jobs by type, including the general job properties.
- From either the list of your jobs or the list of all jobs, you can drill down to details for a specific job by clicking the job name. You can view overview job information, source object processing details, and alerts.

Note: Usually, a job name corresponds to the ingestion task name. For database ingestion jobs, the job name has the format *taskname-job_instance_number*. The number is incremented each time the job is deployed.

Other resources

For more information, visit the Informatica Documentation Portal .

You can also access the latest Mass Ingestion and Administrator help information from within the Informatica Intelligent Cloud Services interface.

Author

Virginia Pfeifle